Abstraction-based branch and bound approach to Q-learning for hybrid optimal control

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Summarising video available at `https://youtu.be/2bc0iHZFf24`

Optimal control problem for hybrid system with affine subsystems.
State-space: $\mathcal{Q} \times \mathcal{X}$ for $\mathcal{X} \subseteq \mathcal{R}^{n_x}$ and finite set $\mathcal{Q}$.

Branch and bound algorithm

$\bar{\beta} \leftarrow \infty$ // No upper bound (UB)
$\mathcal{N} \leftarrow \{\emptyset\}$ // Empty trajectory in priority queue
while $\mathcal{N} \neq \varnothing$ **do**
  $\mathbf{v}_l \leftarrow h(\mathcal{N})$ // Heuristic picks trajectory
  $\mathcal{N} \leftarrow \mathcal{N} \setminus \{\mathbf{v}_l\}$ // Remove from prority queue
  if $\hat{Q}((q_0, x_0), \mathbf{v}_l) \leq \bar{\beta}$ // *branch LB* $\leq$ *global UB*
  then
    for $v' \in \mathcal{V}$ **do**
      $\hat{\mathbf{v}}_{\hat{l}}, \hat{\mathbf{u}}_{\hat{l}}, \hat{\beta} \leftarrow \beta(q_0, x_0, (\mathbf{v}_l; v'))$ // Any feasible solutions ?
      Q-learning for lower bound Q along trajectory $\hat{\mathbf{v}}_{\hat{l}}, \hat{\mathbf{u}}_{\hat{l}}$
      if $\hat{\beta} < \bar{\beta}$ **then**
        $\bar{\mathbf{v}}_{\bar{l}}, \bar{\mathbf{u}}_{\bar{l}}, \bar{\beta} \leftarrow \hat{\mathbf{v}}_{\hat{l}}, \hat{\mathbf{u}}_{\hat{l}}, \hat{\beta}$ // Update UB
        $\mathcal{N} \leftarrow \mathcal{N} \cup \{(\mathbf{v}_l; v')\}$ // Add to priority queue
return $\bar{\mathbf{v}}_{\bar{l}}, \bar{\mathbf{u}}_{\bar{l}}$

Exploration-exploitation trade-off

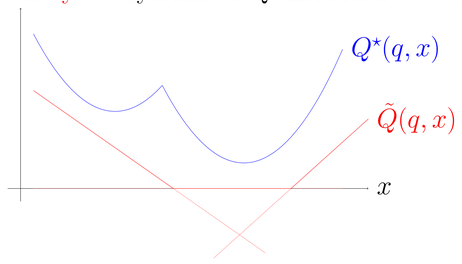
Upper bound β obtained using MPC with Lyapunov function $L(q, x)$. Lyapunov function $L(q, x)$ is zero at the target, finite in some controlled invariant set around the target, infinite otherwise.
The largest the controlled invariant set, the faster the algorithm gets upper bounds.
Which heuristic h should be used ?

Exploration: Priority to lowest lower bound. This needs a Lyapunov function with large controlled invariant set to still get upper bounds early ;

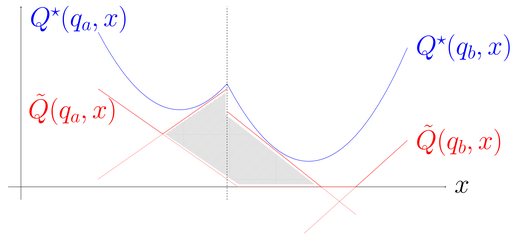
Exploitation: Depth first search, break ties with lower bound. Quickly get upper bounds, even with poor Lyapunov function.

Q-learning

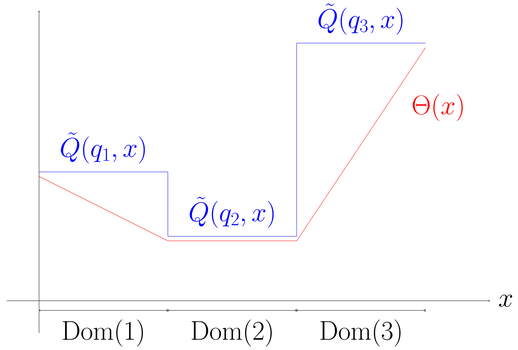
Optimal control of hybrid systems so Q-function is nonconvex.



Gap between polyhedral lower bound can be reduced by increasing horizon [2], or refining the hybrid state-space abstraction:



Dual obtained by solving a strong formulation. Epigraph of the nonconvex $\tilde{Q}(q, x)$ below replaced by its convex hull: the epigraph of $\Theta(x)$ below:

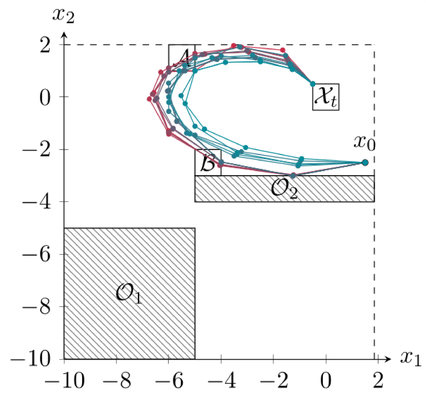


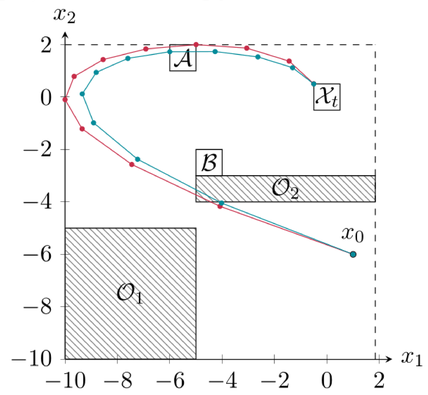
Numerical results

Number of iterations of Branch and bound algorithm:

Bellman-like Q-function	# iterations for $\mathcal{I}_1$	# iterations for $\mathcal{I}_2$
Q_1	197 234	9 388 410
Q_2	1076	96
Q_2 with learning	789	75
Q_2 learning on $\mathcal{I}_1$	782	74
Q_2 learning on $\mathcal{I}_2$	800	74

Feasible solutions for $\mathcal{I}_1$ (left) and $\mathcal{I}_2$ (right) with increasingly better bounds:





The numerical results can seemlessly be reproduced using the codeocean capsule [1]. Algorithms implemented in the Dionysos Julia package.

[1] Benoît Legat, Raphaël M. Jungers, and Jean Bouchat. *Abstraction-Based Branch and Bound Approach to Q-learning for Hybrid Optimal Control*. 2020. arXiv: 2011.11029 [math.OC].

[2] Sandeep Menta et al. "Learning Solutions to Hybrid Control Problems Using Benders Cuts". In: *Proceedings of the 2nd Annual Learning for Dynamics & Control Conference*. 2020.